A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

🟧 Cyclic groups 🟪 Alternating groups 🟦 Chevalley groups 🟩 Steinberg groups 🟨 Suzuki and Ree groups 🟪 Sporadic groups

HOW TO READ A CELL

$\cong A_2(2)$ — other descriptions of the same group

$A_1(7)$ — the group

— prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple, or it is isomorphic to one printed elsewhere — every group appears only under its simplest name:

$A_2(1) \to C_3$
$A_1(4), A_1(5) \to A_5\ [\ A_4(1)\ ]$
$A_1(9) \to A_6\ [\ A_5(1)\ ]$

$A_2(2) \to A_1(7)$
$A_3(2) \to A_8\ [\ A_7(1)\ ]$
${}^2A_3(2) \to B_2(3)$

$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6\ [\ A_5(1)\ ]$
$G_2(2)' \to {}^2A_2(3)$

${}^2G_2(3)' \to A_1(8)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)

${}^2A_n(q) = U_{n+1}(q)$ — Unitary
${}^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4

$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7
$E_8(q)$ — Exceptional E_8
${}^3D_4(q)$ — Triality

${}^2E_6(q)$ — Twisted E_6
${}^2B_2(q)$ — Suzuki
${}^2G_2(q)$ — Ree (small)
${}^2F_4(q)$ — Ree (large)

cyclic
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
rank n
q

q = size of the field

Sporadic groups

Mathieu	Mathieu	Mathieu	Mathieu	Mathieu	Janko	Hall–Janko	Janko	Janko	Higman–Sims	McLaughlin	Held	Rudvalis	Suzuki	O'Nan	Conway	Conway	Conway	Fischer	Fischer	Fischer	Harada–Norton	Lyons	Thompson	Baby Monster	Monster
M_{11}	M_{12}	M_{22}	M_{23}	M_{24}	J_1	J_2	J_3	J_4	HS	McL	He	Ru	Suz	O'N	Co_1	Co_2	Co_3	Fi_{22}	Fi_{23}	Fi_{24}'	HN	Ly	Th	B	M
$2^4\cdot3^2\cdot5\cdot11$	$2^6\cdot3^3\cdot5\cdot11$	$2^7\cdot3^2\cdot5\cdot7\cdot11$	$2^7\cdot3^2\cdot5\cdot7\cdot11\cdot23$	$2^{10}\cdot3^3\cdot5\cdot7\cdot11\cdot23$	$2^3\cdot3\cdot5\cdot7\cdot11\cdot19$	$2^7\cdot3^3\cdot5^2\cdot7$	$2^7\cdot3^5\cdot5\cdot17\cdot19$	$2^{21}\cdot3^3\cdot5\cdot7\cdot11^3\cdot23\cdot29\cdot31\cdot37\cdot43$	$2^9\cdot3^2\cdot5^3\cdot7\cdot11$	$2^7\cdot3^6\cdot5^3\cdot7\cdot11$	$2^{10}\cdot3^3\cdot5^2\cdot7^3\cdot17$	$2^{14}\cdot3^3\cdot5^3\cdot7\cdot13\cdot29$	$2^{13}\cdot3^7\cdot5^2\cdot7\cdot11\cdot13$	$2^9\cdot3^4\cdot5\cdot7^3\cdot11\cdot19\cdot31$	$2^{21}\cdot3^9\cdot5^4\cdot7^2\cdot11\cdot13\cdot23$	$2^{18}\cdot3^6\cdot5^3\cdot7\cdot11\cdot23$	$2^{10}\cdot3^7\cdot5^3\cdot7\cdot11\cdot23$	$2^{17}\cdot3^9\cdot5^2\cdot7\cdot11\cdot13$	$2^{18}\cdot3^{13}\cdot5^2\cdot7\cdot11\cdot13\cdot17\cdot23$	$2^{21}\cdot3^{16}\cdot5^2\cdot7^3\cdot11\cdot13\cdot17\cdot23\cdot29$	$2^{14}\cdot3^6\cdot5^6\cdot7\cdot11\cdot19$	$2^8\cdot3^7\cdot5^6\cdot7\cdot11\cdot31\cdot37\cdot67$	$2^{15}\cdot3^{10}\cdot5^3\cdot7^2\cdot13\cdot19\cdot31$	$2^{41}\cdot3^{13}\cdot5^6\cdot7^2\cdot11\cdot13\cdot17\cdot19\cdot23\cdot31\cdot47$	$2^{46}\cdot3^{20}\cdot5^9\cdot7^6\cdot11^2\cdot13^3\cdot17\cdot19\cdot23\cdot29\cdot31\cdot41\cdot47\cdot59\cdot71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over "the field with one element" a group of Lie type degenerates to its Weyl group, and the cell $A_n(1)$ holds the alternating group A_{n+1}. Only type A is simple there, so that row is exactly the alternating groups.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation, and so does a cell whose factorisation is too long to print.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- ${}^2B_2(q)$ and ${}^2F_4(q)$ exist for $q = 2^{2m+1}$ and ${}^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- ${}^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- A symbol with a subscript and no field is a group over "F_1": C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.